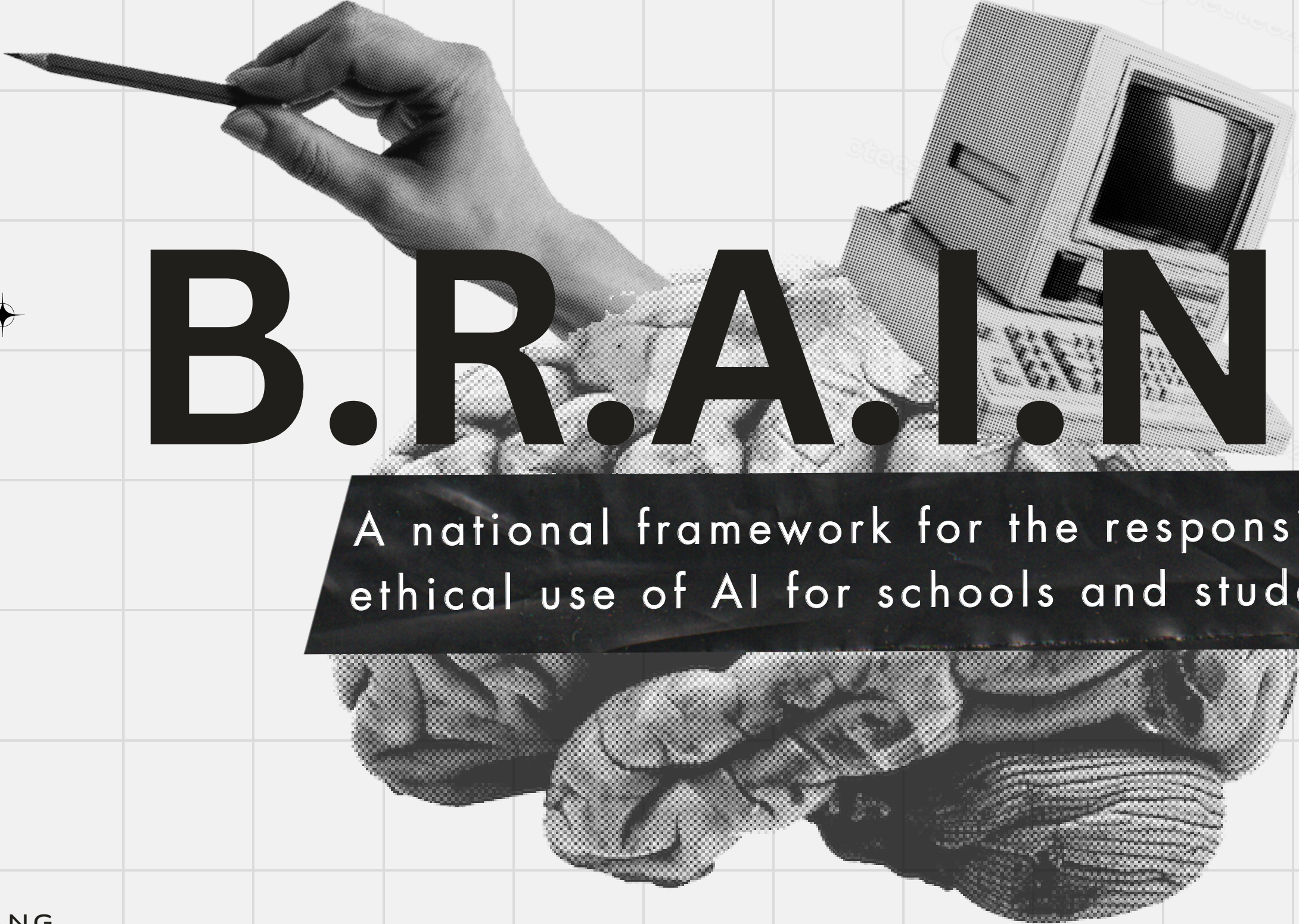
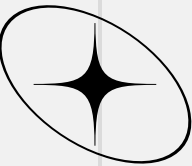




# B.R.A.I.N.S

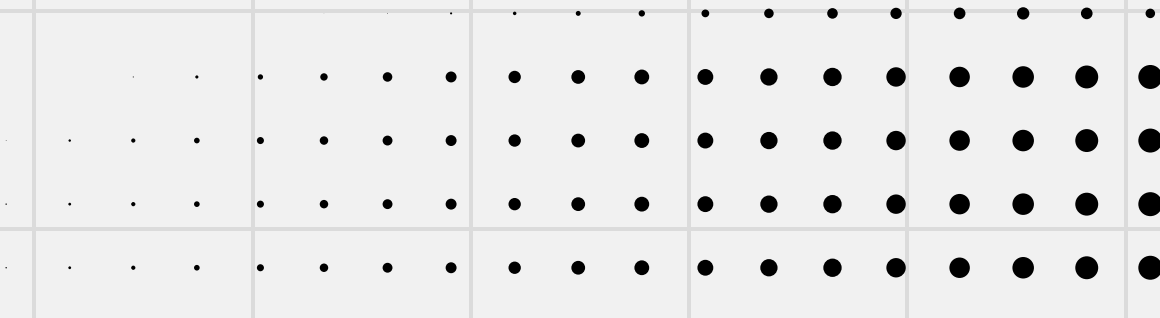
A national framework for the responsible,  
ethical use of AI for schools and students.

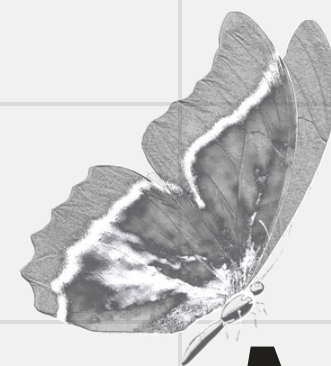
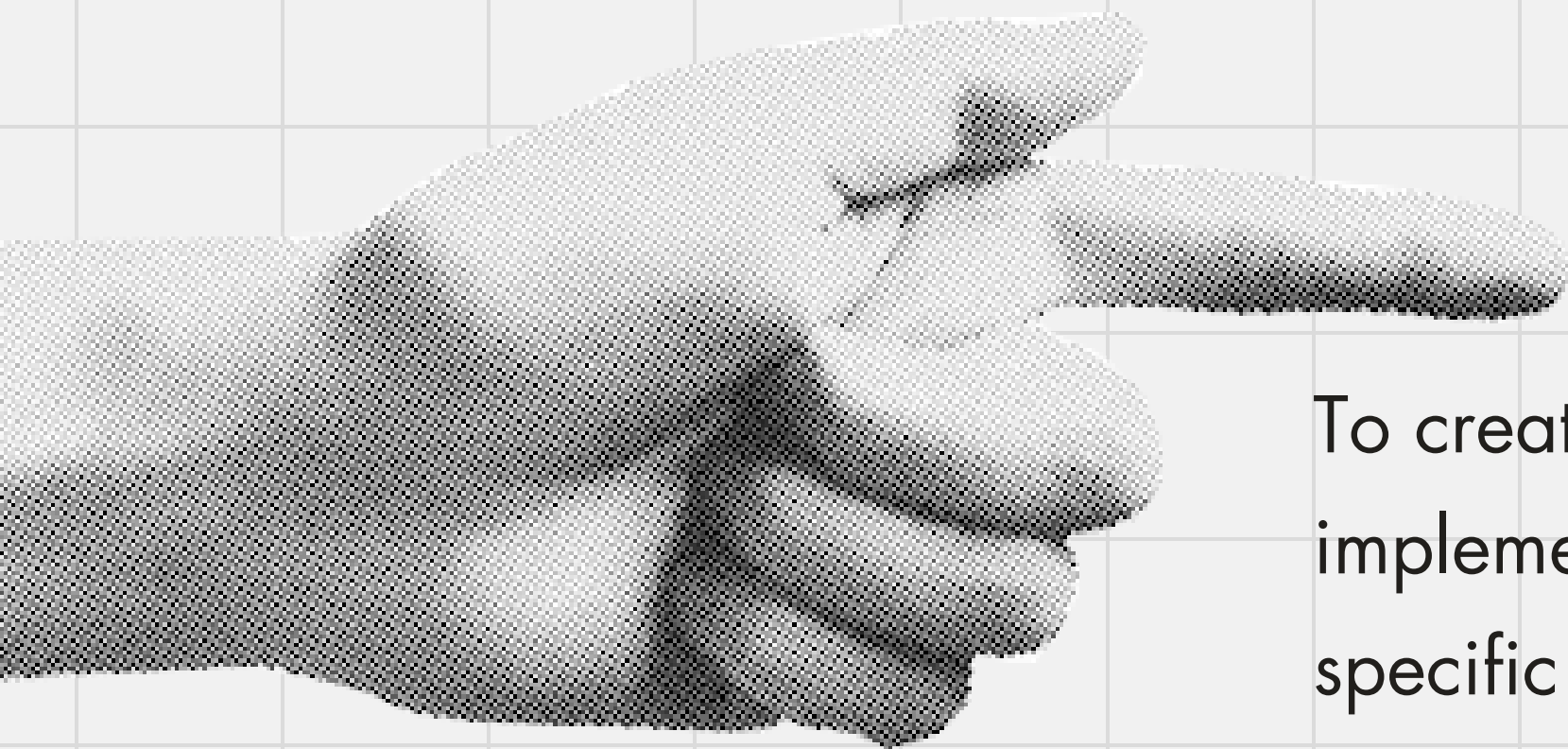
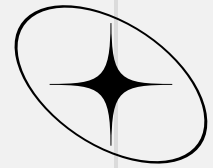
TIFFANY LU  
CHRISTIAN WANG  
ISAAC CHAN  
DAVID CASTRO



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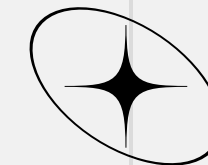
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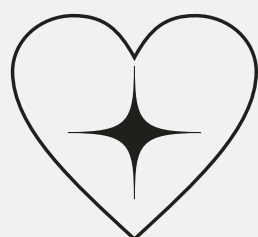
# Mission

To create a national framework for the ethical and responsible implementation of AI technologies in educational settings with specific guidance on how AI can advance, not hinder, the educational goals of schools and students, especially in underserved communities.



# The State of AI

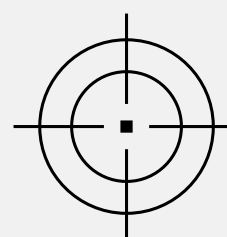
Assessing the current impact of artificial intelligence systems in the educational setting.



## ITS

**Intelligent tutoring systems (ITSs)** are an early application of rule-based expert systems in education; ITSs are used in schools to help teachers accommodate a wide range of student abilities in heterogeneous classrooms—a well-known challenge for many teachers (Murphy, 2019).

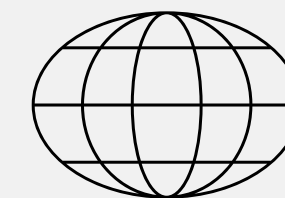
→ ITSs are limited to rule-based AI architecture, thus exclusive to math, physical sciences, etc.



## AES

**Automated essay scoring (AES)** is one of the most mature applications of AI in education; human essay grading is extremely time-intensive, so many teachers are often reluctant to assign extended writing projects (Murphy, 2019).

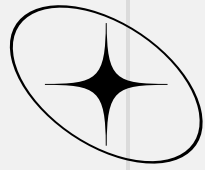
→ *Adversarial input* fools some NLP AES models, however, AES systems and human graders generally give similar scores.



## LLMs/GAI

Although it is hard to currently determine the long-term impact of **Generative AI (GAI)** technologies (like ChatGPT), in an educational setting, GAI technologies have allowed students and faculty alike to accelerate exploration and research with greater accessibility to information (Cornell).

→ LLMs/GAI cannot replace human creativity, judgment, ethics, or responsibility, all of which are essential for learning.



# Current Impact of AI

## *Statistics & usages*

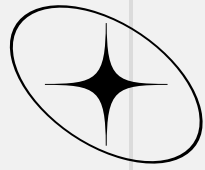
According to a survey from the Digital Education Council, **24%** of students globally **use AI daily**, and **54%** use it **at least weekly**. ChatGPT is the most commonly used AI tool, reported by 66% of respondents ([Kelly, 24](#)). In June 2023, the ACT conducted a survey of 4006 students from grades 10 to 12, which showed that **46% of students use AI tools for school assignments**. The main reasons for students not using them were lack of interest, trust, and not knowing enough about it. Although less than half of students use AI tools for school, these tools have a significant impact on students' academic achievement. **53% of students with ACT composite scores in the top quarter used AI tools**, compared with 36% of those in the bottom quarter ([ACT, 23](#)).

## *Regulations?*

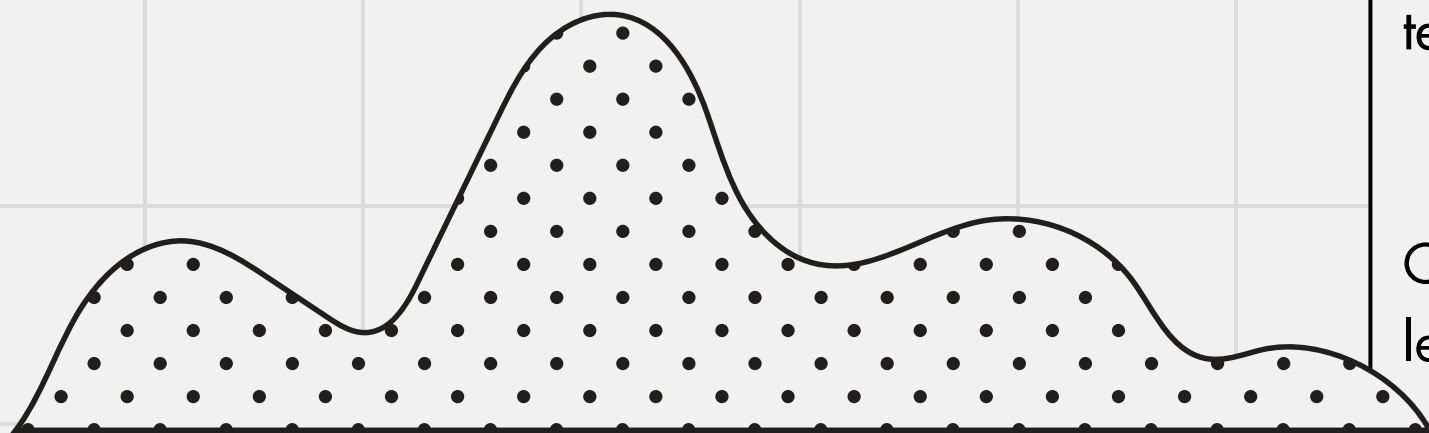
- 58% of students don't feel like they have enough knowledge of usage in AI
- 48% of students feel like they are not prepared for an AI-enabled workforce
- 80% of students thinks that the current university's integration of AI tools is not what they expected

Survey from [Digital Education Council](#), a global alliance of universities and industry representatives focused on education innovation

Most students feel **unprepared** to use AI and **underwhelmed** by the current regulations.



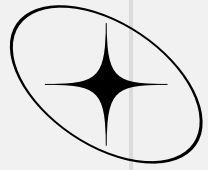
# Existing Efforts for Ethical AI Use in Education



MIT has developed an AI ethics curriculum in their **Dalai Lama Center for Ethics and Transformative Values** that involves students critically analyzing algorithms used in real-world applications like YouTube's recommendation system, exploring both how these systems work and the ethical dilemmas they present. This curriculum emphasizes the importance of understanding the mechanics of AI to discuss its ethical implications, which aligns with the idea of teaching how AI works as a foundation for understanding its ethics. ([Payne, August 2019](#))

Additionally, computer science education is increasingly focusing on responsible AI usage, with educators advocating for teaching the ethics and impact of AI at various levels, including high school. Some organizations, like the **Computer Science Teachers Association**, support initiatives to integrate computational thinking and AI ethics from a young age, showing growing interest in blending AI ethics with technical education. ([Shein, July 2024](#))

Overall, the existing efforts for Ethical AI use involve educating the public or at least a student body about how AI works.



# Some Ethical Concerns on the Impact of Artificial Intelligence

✦ SOLVATHON ✦

2024



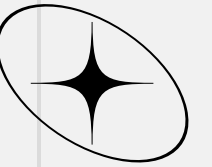
## *Data Privacy & Security*

AI systems in education often rely on collecting and analyzing large amounts of data about students. This raises questions about how this data is stored, who has access to it, and how it is used. There are risks of breaches of privacy and potential misuse of data (Trust, 2023).

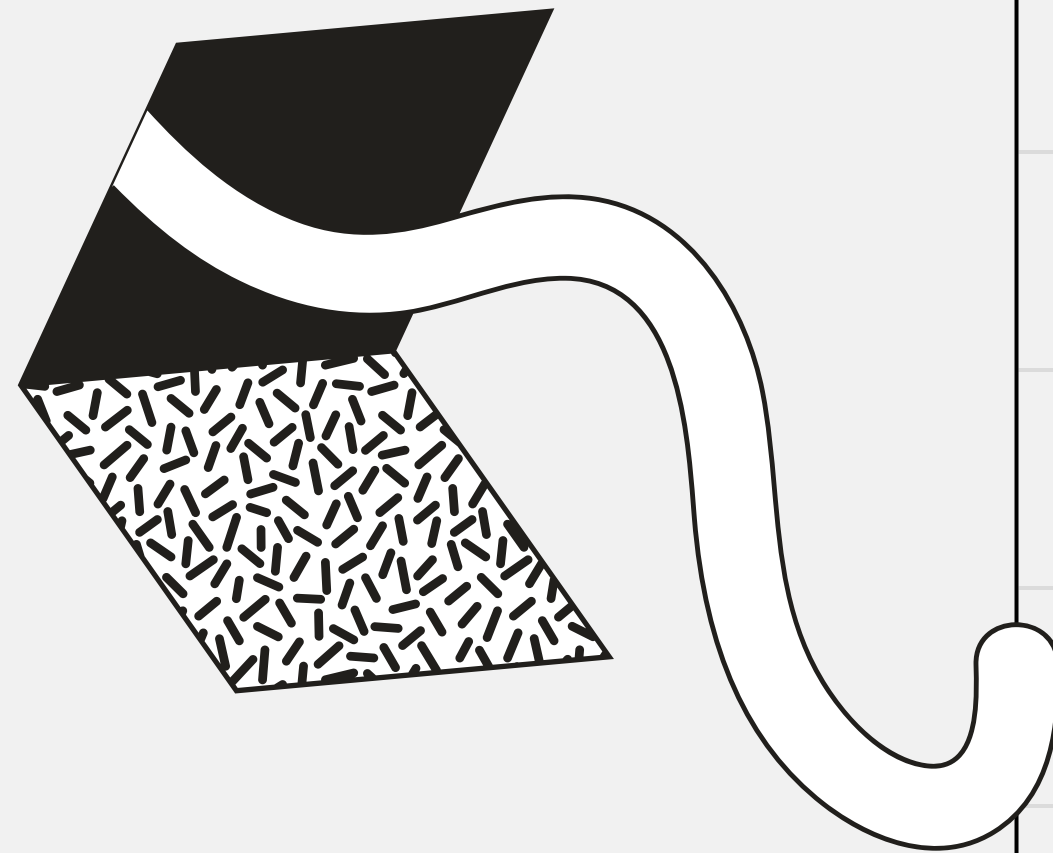
## *Technology Encodes Biases*

Educational AI can be subject to bias, depending on its training data. E.g. based on the data it was trained with, an AI tutoring system could favor the responses of certain types of students over others per systematized sociocultural and socioeconomic norms. Through the enforcement of systemic biases and perspectives as '*truth*,' AI could perpetuate existing biases and inequities in broader social and educational spheres.





# Some Ethical Concerns on the Impact of Artificial Intelligence

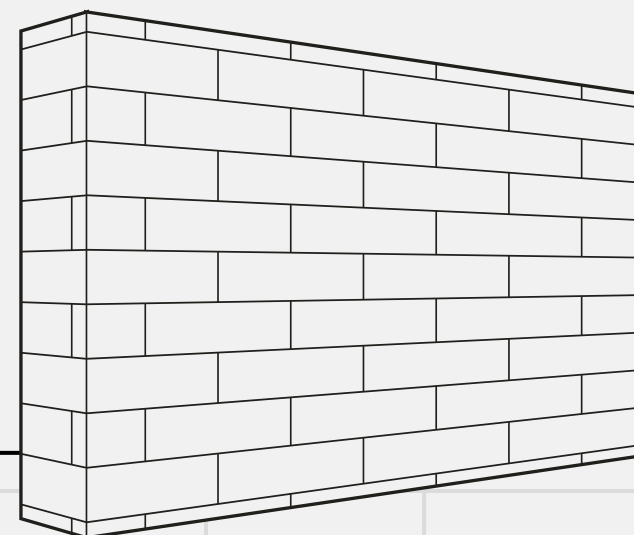


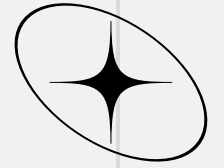
## *Plagiarism*

As Generative AI models rely largely on human-produced training data sets, some ethicists have deemed the usage of GAI technologies as plagiarism, thus unacceptable in an academic setting. In addition, plagiarism of any kind devalues the learning of students who partake in it.

## *Inequity of Access*

Although AI is a powerful and generally accessible tool, school districts with lesser funding, especially in areas where families may not be technologically literate or have the financial means to access AI (e.g. they lack a stable internet connection or cannot afford hardware capable of running modern AI-powered applications), will fall behind other, wealthier areas, furthering inequities in education. For a successful program to be implemented, inequity of access to AI technologies and supplementary technologies, like computers and internet, must also be considered.





# Our Take On Ethical AI

## Our Goals: **EEE**

### EDUCATE

- **AI Plagiarism:** One core ethical issue with AI is its potential to reproduce data without proper credit, essentially engaging in unintentional plagiarism. Teaching students how AI models are trained on large datasets will expose the lack of transparency in data sourcing, thus fostering awareness
- **Consent and Understanding:** The goal here is to educate students and schools on responsibly engaging with AI. By understanding the implications of using AI, users can make informed, consensual choices about what data they share and how AI utilizes it.

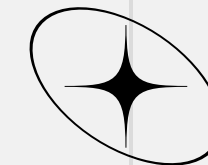
### EMPOWER

**Broadening Access:** Current AI educational frameworks often focus on higher education, which limits secondary school students' access to vital AI knowledge. Expanding this to secondary schools ensures that a broader segment of students gains the technical and ethical skills needed to navigate an AI-driven world.

**Nationwide Impact:** Empowering both students and educators nationwide will create a ripple effect, where students from diverse backgrounds can interact with cutting-edge technology and learn how to harness it ethically and responsibly.

### ENGAGE

- **Bridging the Gap:** There is a significant knowledge gap regarding AI between high school students and undergraduates. Your co-op model closes this gap by creating opportunities for undergraduate students to mentor and teach secondary students.
- **Addressing Oversaturation:** Many computer science graduates face stiff competition in the job market. Engaging them in educational roles allows them to apply their knowledge in meaningful ways, diversifying their career prospects while simultaneously addressing ethical AI issues.

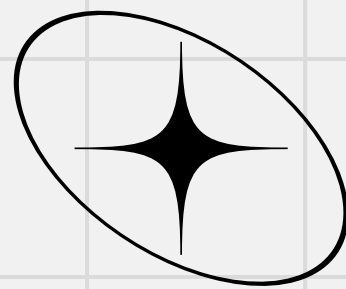


# How will we attain **EEE?**



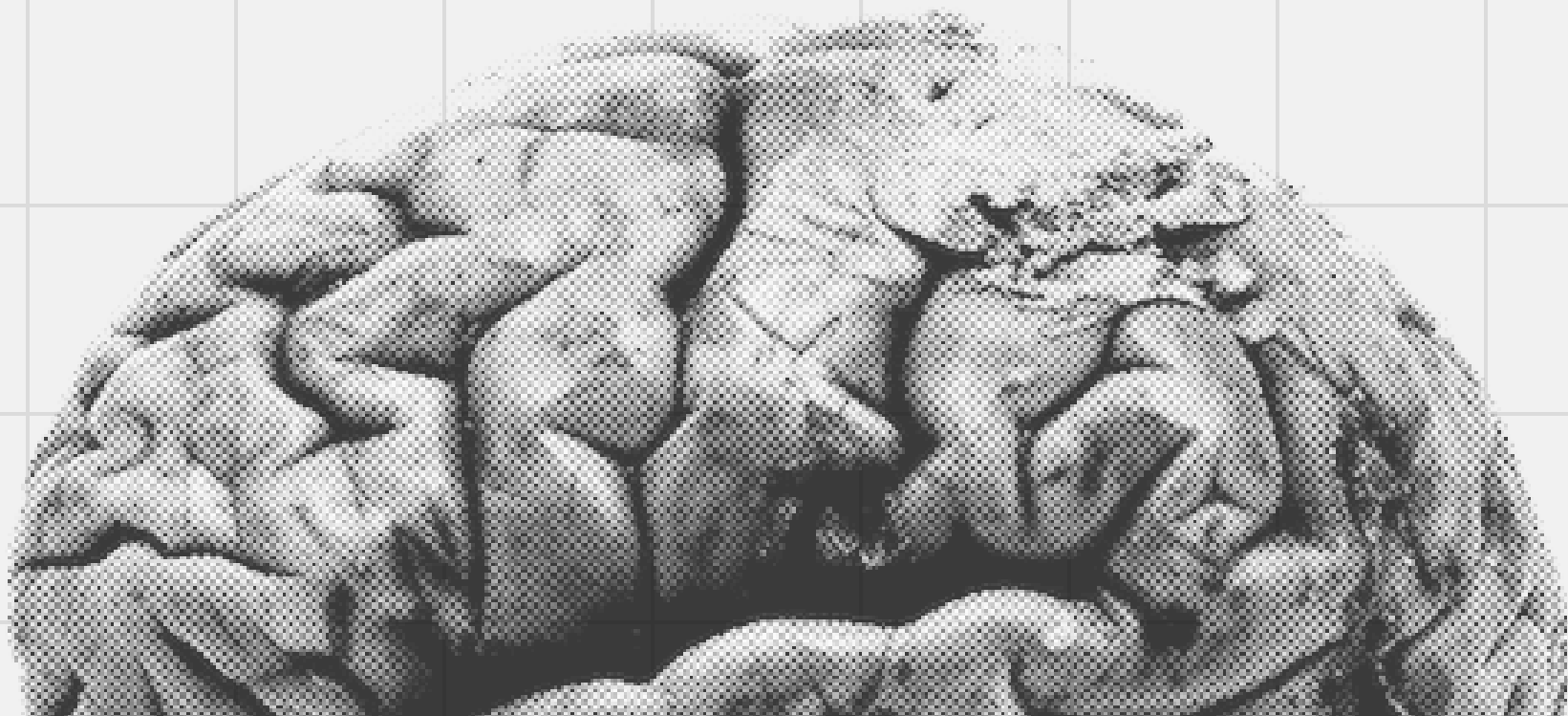
INTRODUCING...

OUR FRAMEWORK:



# B.R.A.I.N.S

Budget for Responsible AI Integration in National Schools



## **B: Building Ethical Foundations (\$210,000)**

Focus on stakeholder engagement, curriculum development, and training programs to lay the groundwork for responsible AI use in education.

## **A: AI Equity Assurance (\$300,000)**

Develop and launch tools and studies to ensure equitable, bias-free AI integration, focusing on long-term impacts.

## **N: Networking and Outreach (Public Awareness) (\$180,000)**

Spread the word! Use public campaigns to inform and engage stakeholders on the benefits and risks of AI in education.

# B

# R

# A

# I

# N

# S

## **R: Resource Empowerment (\$230,000)**

Empower students and teachers with AI-enhanced tools and knowledge-sharing platforms.

## **I: Implementation & Impact Measurement**

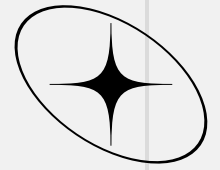
Guide schools with implementation guidelines and maintain oversight to ensure responsible, efficient use of AI resources.

## **S: Safety Net (Contingency) (\$80,000)**

A 10% buffer to cover unforeseen expenses and keep the program on track.

*BRAINS: using our collective brainpower to integrate AI into education intelligently, ethically, and equitably.*

# Phase 1: Building Ethical Foundations (Months 1-2)



*Objective: Lay the foundation for AI ethics education in schools.*

## Stakeholder Engagement:

**Cost: \$50,000**

Collaborate with schools, universities, and tech companies to organize stakeholder discussions on AI ethics. Partner with state Departments of Education to co-host events, reducing expenses. Leverage virtual platforms to minimize travel costs and involve a broad range of experts.

## Curriculum Development:

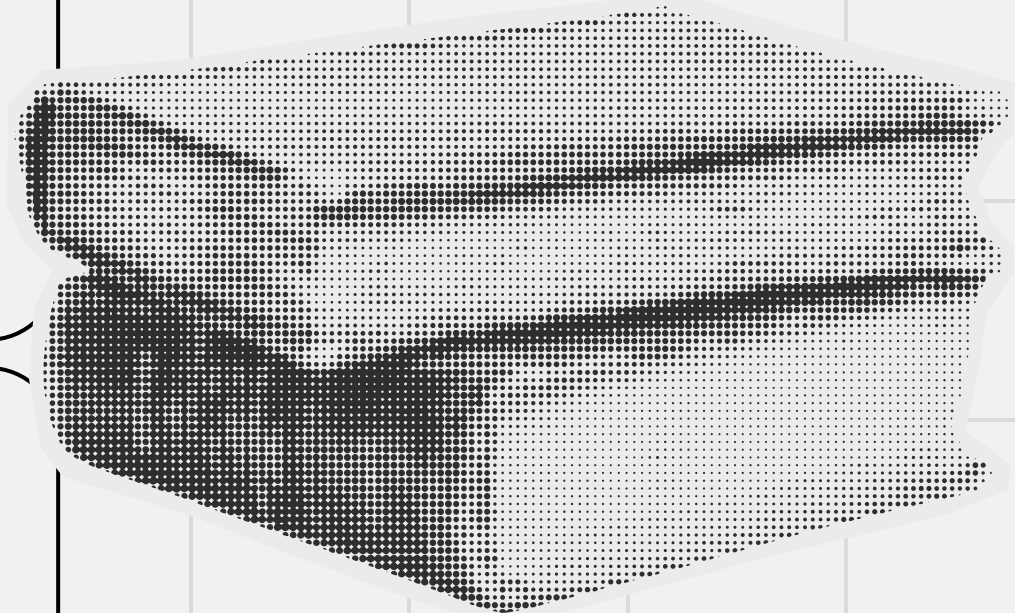
**Cost: \$100,000**

Work with AI and educational experts to create a comprehensive curriculum that teaches secondary students how AI works, ethical concerns, and responsible use. Partnering with tech companies and universities helps reduce costs by using existing educational materials and expertise.

## Undergraduate Training Program:

**Cost: \$60,000**

Recruit undergraduate co-op students through partnerships with NSF and education NGOs to deliver the AI ethics curriculum in schools. This low-cost approach brings in motivated, trained undergraduates while providing valuable hands-on teaching experience for them.



# Phase 2: Resource Empowerment (Months 3-4)

*Objective: Equip students and educators with AI literacy and resources.*

## Co-op Program Launch:

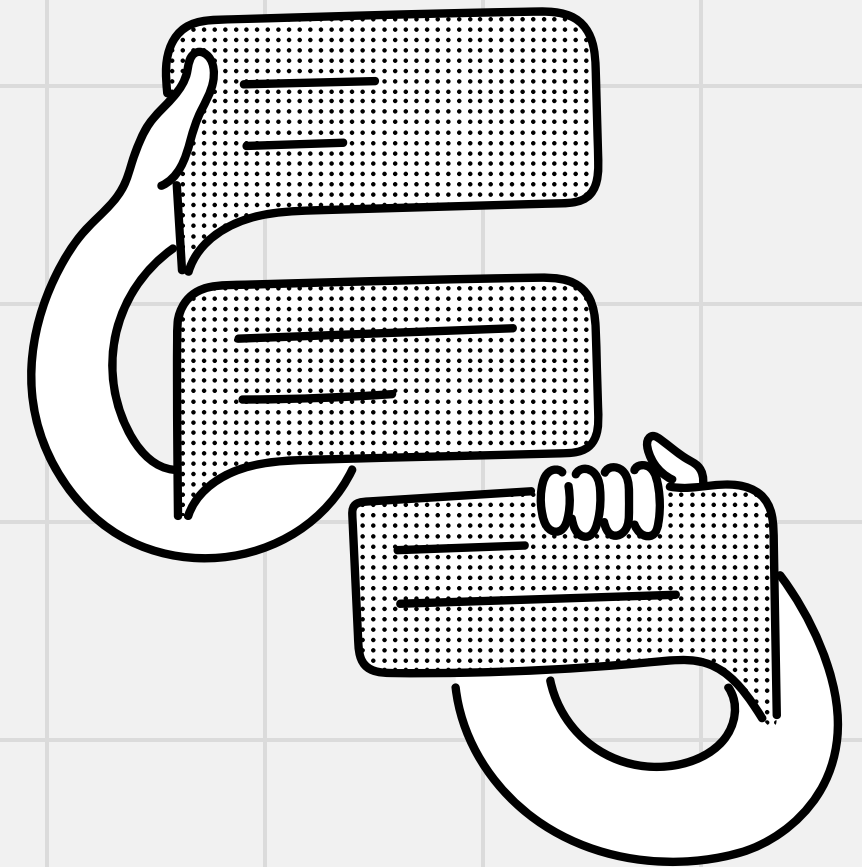
**Cost: \$150,000**

Implement the co-op program where undergraduate students teach AI ethics in schools. Initially, the focus should be on urban areas for cost-effective scaling, and public transportation networks should be used to minimize logistical expenses. Broaden access by hosting virtual workshops, allowing nationwide reach without additional infrastructure.

## AI-Enhanced Lesson Plan Repository:

**Cost: \$80,000**

Develop a centralized, cloud-based repository where teachers can access and share AI-augmented lesson plans. Use low-cost cloud services like Google or Microsoft, and encourage teachers to contribute voluntarily, reducing ongoing content development costs. OET will vet submissions for quality and alignment with national standards.



# Phase 3: **AI** Equity Assurance (Months 5-7)

*Objective: Ensure equitable access to AI tools and evaluate program success.*

## AI Equity Audit Tool:

**Cost: \$100,000**

Work with the Office for Civil Rights (OCR) and tech partners to develop an AI Equity Audit Tool, ensuring AI tools are free from bias and accessible to all students. Using open-source technology will help reduce costs while adapting the tool specifically for educational use.

## Longitudinal Study on AI in Education:

**Cost: \$200,000**

Partner with the Institute of Education Sciences (IES) to conduct a detailed study on AI's long-term impact on student outcomes, such as critical thinking and knowledge retention. This will guide future AI policies and ensure the program's success. Matching funds or partnerships with research institutions can reduce overall cost.



# Networking and Oversight

*How can we make sure this runs successfully?*

## Public Awareness Campaign:

**Cost: \$60,000**

Engage in public outreach through social media, education-focused media outlets, and NGOs to inform and educate students, teachers, and parents about AI ethics and responsible usage in schools. Leverage low-cost media and existing outreach channels for broad impact.

## Guidelines for State & Local Implementation:

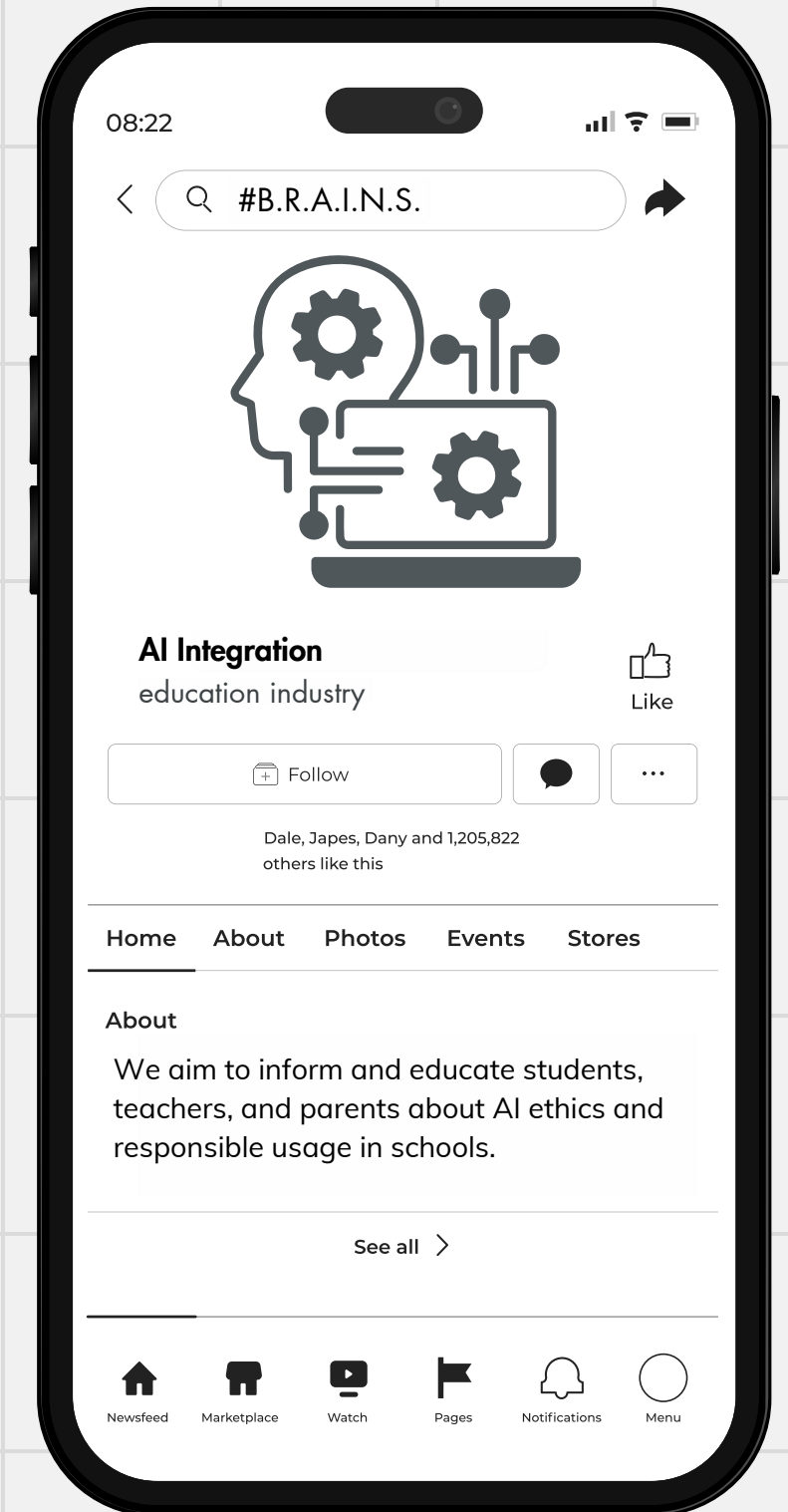
**Cost: \$40,000**

Develop clear, practical guidelines for state and local education agencies on implementing AI tools. Distribute these guidelines digitally to minimize costs and host low-cost webinars for school administrators and educators.

## Oversight Mechanism Setup:

**Cost: \$80,000**

Establish a small but effective oversight team within the Department of Education to monitor the program, collect data, and adjust based on feedback. Automating data collection and utilizing existing infrastructure will keep operational costs low.



# Safety Net

## Contingency

### Cost: \$80,000

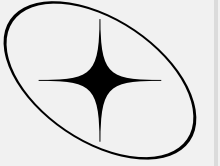
Lastly, a 10% contingency buffer will be kept to cover any unexpected expenses and ensure the program remains on track, regardless of unforeseen challenges.

# Success Metrics

Key metrics for measuring the program's success beyond implementation:

- **Student Learning Outcomes:** Assess the increase in knowledge of AI ethics among secondary students through pre- and post-program surveys and quizzes.
- **Teacher and Administrator Feedback:** Measure how comfortable secondary school teachers feel incorporating AI ethics into their own curricula after the co-op students leave.
- **Career Development for Co-op Participants:** Track the career progression of co-op students. Have they secured jobs in AI or education after the program? This could measure how well the program bridges the knowledge-to-job market gap.
- **National Reach:** Long-term, measure how many secondary schools implement similar programs across the country and track the number of undergraduates participating in AI education outreach.

# Possible Constrains



## Urban/Rural & Technological Infrastructure Gaps

Rural schools often lack the technological infrastructure (high-speed internet, hardware) necessary for AI tools, while urban schools may have the resources but face overcrowding and underfunding. Rural areas, in particular, may also struggle to attract and retain undergraduate co-op participants.

### Safety Net

The program will allocate extra resources to tech infrastructure in rural areas (e.g. subsidized high-speed internet). It will also create virtual co-op opportunities (Zoom, Google Meet).

## Teacher Workload and Resistance

Teachers are often overburdened, especially in underfunded schools, and may resist learning new AI tools. The fear of AI replacing their role, particularly in areas such as grading or tutoring, could create some reluctance towards the program.

### Safety Net

The program will include professional development for educators; AI tools will be positioned as assistive to reduce workload rather than a replacement for human, teacher roles.

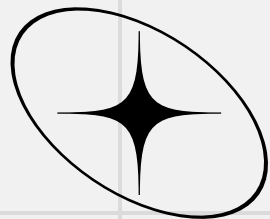
## Privacy Concerns

Parents, especially in marginalized communities, may mistrust AI tools due to concerns regarding data privacy, surveillance, and misuse of their child's information. There is also some concern, similar to Teacher Workload and Resistance, regarding AI overtaking human judgement in education.

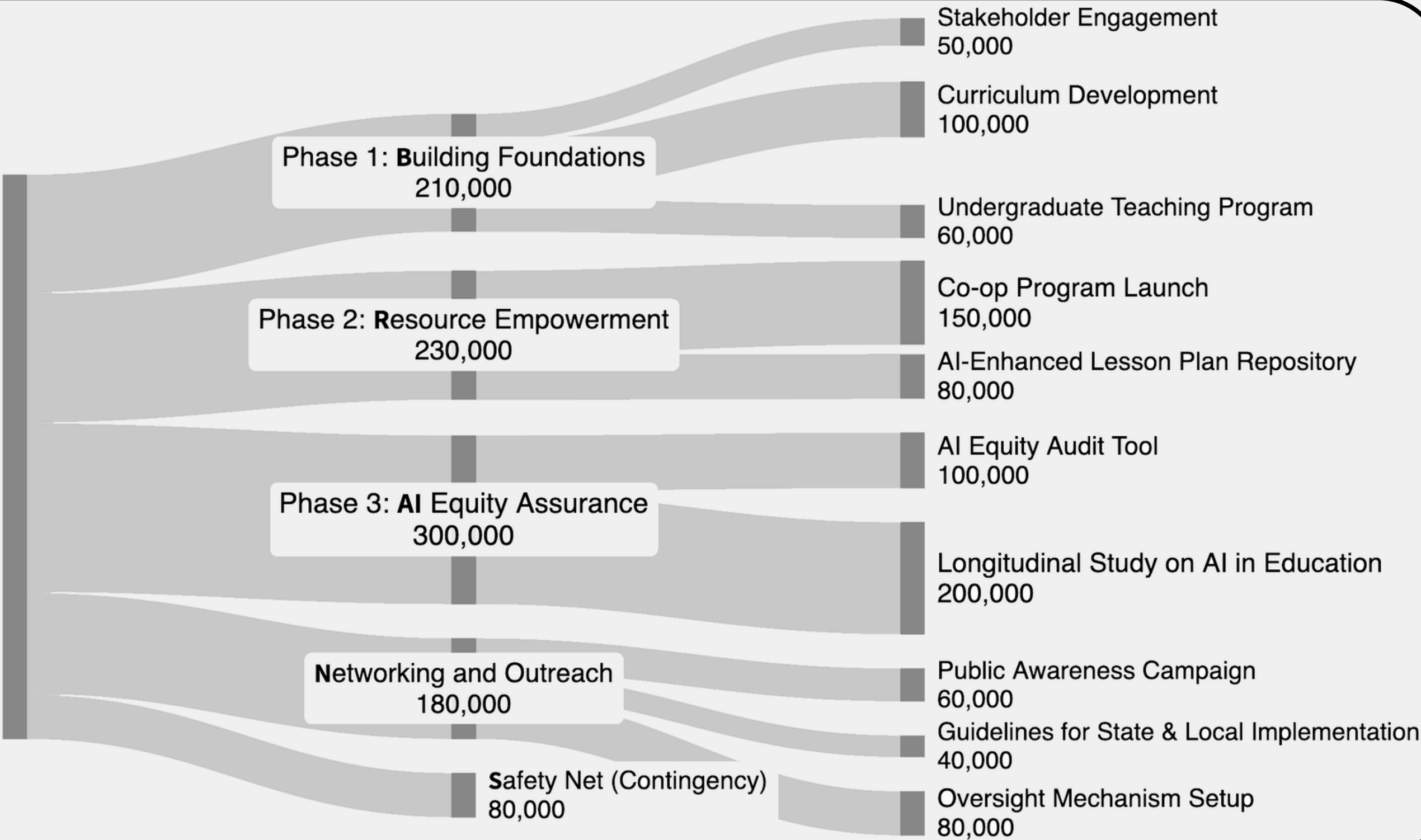
### Safety Net

Partnering with the Office for Civil Rights (OCR) ensures AI tools undergo equity audits, addressing privacy and fairness concerns. Privacy protocols will be integrated within AI systems; data will be anonymized.

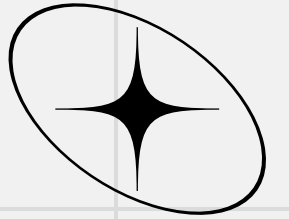
# B.R.A.I.N.S Budget Breakdown



**BRAINS Budget**  
1,000,000



# Strategic Partnerships and Cost Optimization



**National Institute of Standards and Technology (NIST):** Development of AI standards and benchmarks for educational tools.

**Office of Educational Technology (OET):** Coordination of AI initiatives in line with national technology policies.

**National Science Foundation (NSF):** Grant support and research on AI-driven personalized learning.

**Institute of Education Sciences (IES):** Evaluation and research on the impact of AI on learning outcomes.

**Office for Civil Rights (OCR):** Ensuring AI tools are equitable and accessible for all students.

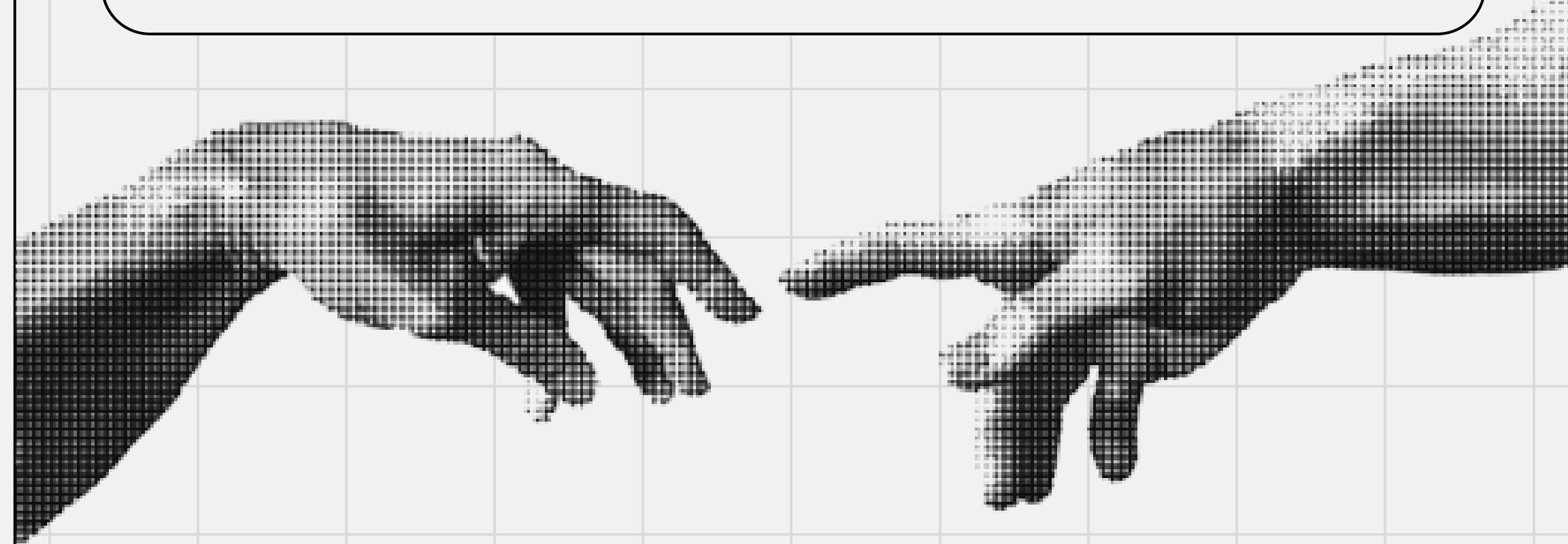
**Non-Governmental Organizations (NGOs):** Engage educational NGOs to develop content and provide additional funding or expertise: 501(c)(3) aiEDU, Team4Tech, Educational Testing Service (ETS)

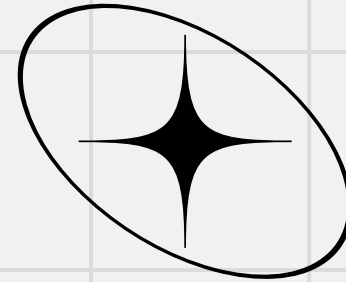
# Intersectional Impacts of the **B.R.A.I.N.S** Framework

Through the **B.R.A.I.N.S** Framework, and specifically through programs the **co-op program** and **AI-enhanced lesson plan repository**, we can ensure AI-powered educational resources are distributed accessibly and equitably, allowing previously underserved and underresourced communities to have the same level of access to epistemic resources as privileged communities.

→ Granting greater accessibility to epistemic resources via AI in underserved areas can help close gaps in knowledge, wealth, and standard of living as students are given a greater opportunity to reach the same educational benchmarks as other regions, consequently driving up the number of college-educated adults and the rate of innovation in the nation as a whole.

Additionally, creating tools like the **AI Equity Audit Tool**, we provide an institutional check on the implicit biases of AI not only in the educational setting but of AI as a whole. In both the educational and broader social spheres, institutionalized biases that often are pushed onto students during formative years can be diminished over time through these tools, combating common prejudiced stereotypes in America.

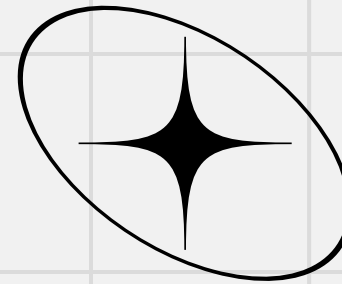




# B.R.A.I.N.S

The future for innovative, ethical, and equitable education.





# Thank You

Sincerely,  
Tiffany, Christian, Isaac, and David

